UNIT 5

Tree

1) Which node in a tree has no parent?
   A) Leaf Node
   B) Root Node
   C) Sibling Node
   D) Internal Node
2) Explain the concept of tree traversal and describe its different types.
3) Construct a Binary Search Tree by inserting the following elements in the given order:
   40, 20, 60, 10, 30, 50, 70**.**
   Apply the three standard tree traversal techniques and write the resulting sequences. 4) Apply the Binary Search Tree (BST) insertion algorithm to construct a BST using the following elements: 50, 30, 70, 20, 40, 60, 80 Then apply Inorder, Preorder, and Postorder traversals to the constructed tree.

Graph

1) In graph terminology, an edge that has a specified direction from one vertex to another is called:
   a) A) Undirected Edge
   b) B) Directed Edge
   c) C) Adjacent Node
   d) D) Successor Node
2) Analyze the differences between Adjacency Matrix and Adjacency List representations of a graph with suitable examples.

Hashing

1) Explain the basic concepts of hashing and graphs. Describe any 1 collision resolution techniques in hashing
2) Apply the hash function h(k) = k mod 10 to insert the following keys into a hash table of size 10 using linear probing**:**
   23, 43, 13, 27, 37**.**
   Show the final hash table and indicate the positions where collisions occur.
3) Which hashing technique uses an additional hash function to determine the next position when a collision occurs?
   A) Cuckoo Hashing
   B) Tree Traversal
   C) Adjacency Matrix
   D) Linear Search